

Lab 6: Import, Validate and Export Weather Data

Level: Intermediate Mapping: LO3 | K6 | A2 | A5 | A9 Concepts: paths, CSV/Excel import, validation, export

Scenario

A public-environment team needs a defensible May ozone summary.

Goal

A validated weather dataset and audit-ready summary export.

Files in this folder

- lab-06.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Open the workbook and note the data contract.
2. Import Input_Data with readxl using a project-relative path.
3. Validate required columns, month values and numeric ozone type.
4. Filter May records and calculate the average ozone with missing values handled.
5. Export analysis_output.csv and update the Evidence sheet.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-06.R and read the header, inputs and validation rules.
3. Run the complete script with source("lab-06.R").
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run source("verify.R") and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Validation stops the script if a required column is absent.
- [] May average uses only non-missing ozone records.
- [] Export contains source row count and analysis timestamp.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.